

L3.1 - Intro to Clustering

Clustering:

partition data into k -clustering

Basically k boxes, all points are sorted into one of the boxes.

→ n points, k clusters

n^k possibilities

Given a cluster assignment, how good is it?

→ sum function

$$F(z_1, \dots, z_n) = \sum_{i=1}^n \|x_i - \mu_{z_i}\|_2^2$$

$\mu_i \rightarrow$ average of z_i 'th cluster

$$\mu_k = \frac{\sum_{i=1}^n x_i \mathbb{1}(z_i = k)}{\sum_{i=1}^n \mathbb{1}(z_i = k)}$$

$$\sum_{i=1}^n \mathbb{1}(z_i = k).$$

$$\mathbb{1}(a) = \begin{cases} 1 & \text{if true} \\ 0 & \text{otherwise} \end{cases}$$

→ NP-hard problem.

→ Use heuristic algorithms

L3.2 - Lloyd's Algorithm

Step - I Initializing

$$z_1^0, z_2^0, \dots, z_n^0.$$

0 → iteration

→ Basically runs until convergence.

Step - II

→ Compute means.

$$\forall k \quad \mu_k^t = \frac{\sum_{i=1}^n x_i \mathbb{1}(z_i^t = k)}{\sum_{i=1}^n \mathbb{1}(z_i^t = k)}$$

Step - III Reassignment

$$\forall i \quad z_i^{t+1} = \underset{R}{\operatorname{argmin}} \|x_i - \mu_k^t\|_2^2$$

→ assignment of i^{th} data point to the closest cluster mean

→ If the mean of current assignment is smallest then don't reassign

Step - IV

Convergence.

- First check if converges.
→ Cannot converge if stuck in loop
- No changes for means for all points

★ **FACT** • Lloyd's algorithm converges

- But may not be optimal
- Produces reasonable clusters in practice

Questions

1) Convergence?

2) Is Lloyd's algorithm optimal?

- 1) Convergence.
- 2) Nature of clusters?
- 3) Initialization.
- 4) Choice of k .

L3.3 - Convergence of Kmeans

1) Convergence.

why Lloyd's algorithm converges.

Proof:

Fact I: Let $x_1, x_2, \dots, x_l \in \mathbb{R}^d$

$$v^* = \underset{v \in \mathbb{R}^d}{\operatorname{argmin}} \frac{1}{l} \sum_{i=1}^l \|x_i - v\|^2$$

→ Taking as function of v / Here it means
 derivative $\xi = 0$ / That if you calculate
 distance of all points
 from one single point, the distance v^* is
 lowest when the point is the mean.

$$v^* = \frac{1}{l} \sum_{i=1}^n x_{ij}$$

Say at iteration t

say at iteration t

→ current assignment

$z_1^t, z_2^t, \dots, z_n^t \leftarrow$ clusters

$\mu_k^t \leftarrow$ mean of cluster k
in iteration t

→ algorithm does reassignment

→ $t+1$ iteration

$z_1^{t+1}, z_2^{t+1}, \dots, z_n^{t+1}$

→ Here trying to understand if
this new partition better
than the previous one. How?

$$\sum_{i=1}^n \|x_i - \mu_{z_i^{t+1}}^t\|^2 \quad \text{--- (I)}$$

→ avg distance of all
points from their means.

@ $t+1$

→ The μ^t here means mean at iteration t

$$\sum_{i=1}^n \|x_i - \mu_{z_i^t}^t\|^2 \quad \text{--- (II)}$$

@ t

① is always lower than ②

in case that any points actually change

How is it changing after reassignment

$$\sum_{i=1}^n \overset{\textcircled{I}}{\|x_i - u_{z_i^{t+1}}^{t+1}\|} \quad \text{vs} \quad \sum_{i=1}^n \overset{\textcircled{II}}{\|x_i - u_{z_i^t}^t\|}$$

① $\leq$ ②, why? Using Fact ①

$$\sum_{i=1}^n \sum_{k=1}^K \|x_i - u_k^{t+1}\| \mathbb{1}(z_i^{t+1} = k)$$

★ Basically proved that every iteration, the objective function strictly reduces.

But why is this convergence?

→ The finiteness of number of assignments

→ Objective function reduces & thus monotonically.

strictly monotonically.

→ Therefore algo must converge.
(Doesn't necessarily mean the actual best).

→ No loops in Algo.

2.3.4 - Nature Of Clusters

Assume,

$K=2$, produces 2 clusters
with means μ_1 & μ_2

what can we say about points
assigned into clusters.

By algorithm's construction:-

→ For points in cluster 1

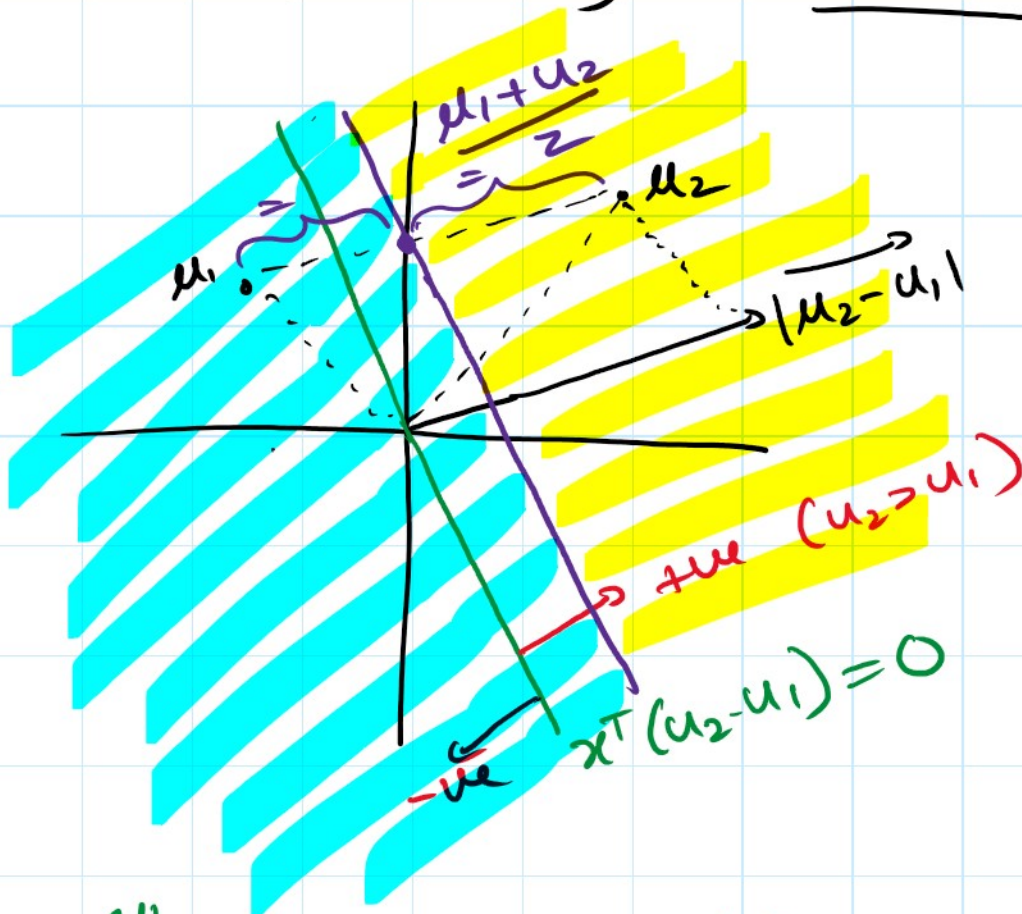
$$\|x - \mu_1\|^2 \leq \|x - \mu_2\|^2$$

$$= \|x\|^2 + \|\mu_1\|^2 - 2x^T \mu_1$$

$$\leq \|x\|^2 + \|\mu_2\|^2 - 2x^T \mu_2$$

$$x^T(u_2 - u_1) \leq \frac{\|u_2\|^2 - \|u_1\|^2}{2}$$

$$x^T(u_2 - u_1) \leq \frac{\|u_2\|^2 - \|u_1\|^2}{2}$$



The area falls depends on R.V.S.

if $\|u_2\| > \|u_1\|$, $\rightarrow$ true
else $\rightarrow$ -ve.

here assumed true.

line position

z_2 .

acc to eq $<$ line so z_1

To find how much the line has

To find how much the line has moved, we need to find a point on it

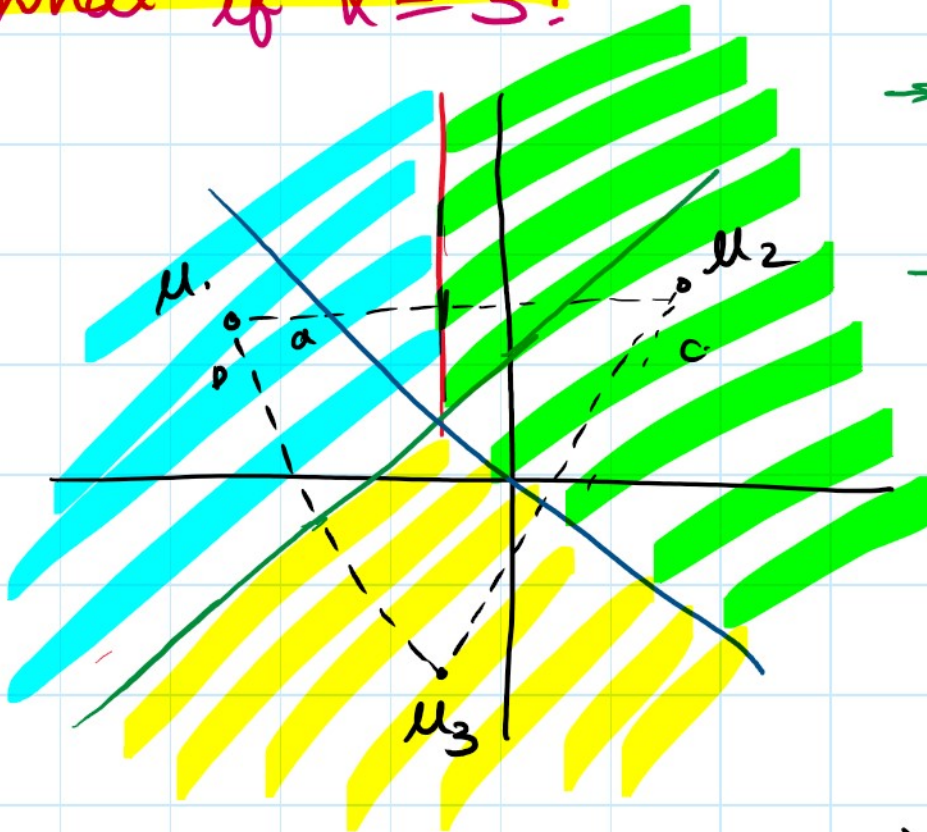
assume $\alpha = \frac{\mu_1 + \mu_2}{2}$

$$\Rightarrow \left(\frac{\mu_1 + \mu_2}{2} \right)^T (\mu_2 - \mu_1)$$

$$= \frac{\|u_2\|^2 - \|u_1\|^2}{2}$$

- Assume avg of means.

What if $k=3$?



→ Draw lines
a, b, c

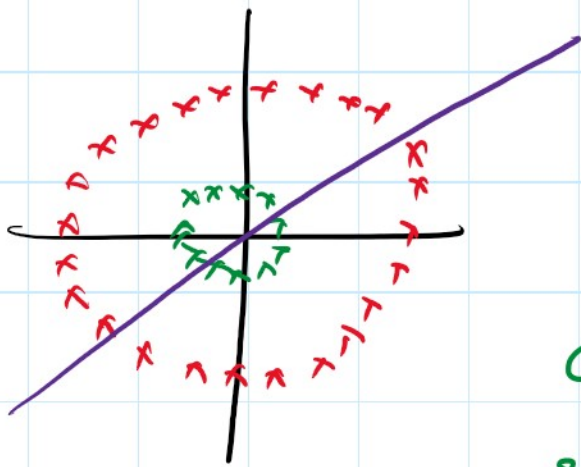
→ bisect

A cluster regions are intersections of:

* Cluster regions are intersections of half spaces. AKA Voronoi regions

Interesting to know:

If 2 concentric circles in different clusters, then line may appear.



→ wrong in this.

→ Data without any clear voronoi structures does not

work with Lloyd's

How To Fix?

→ Kernelize k-means.

→ other SPECTRAL clustering

L3.5 - Initialization of centroids, kmeans++

Initialization

Initialization

→ can do in any way but any ways
to initialize better to have better outcomes
Random.

→ Select randomly, • not mindful,
• might get stuck in local
minimas

Uniformly

→ Uniformly at data from
dataset

→ Run multiple times

K-means++

→ with Better initialization

→ means are as away from each
other as possible

→ do it with probabilities

Steps:

Steps:

→ Choose first mean μ^0 , uniformly at random from $\{x_1, \dots, x_n\}$

→ For $l = \{2 \dots k\}$

→ Choose μ_l^0 probabilistically proportional to score.

→ probabilistically pick a point according to the score we assign a point

→ score assigned based on distance to all previous means, further better

$$\forall x. s(x) = \min_{j=1 \dots l-1} \|x - \mu_j^0\|^2$$

Eg:-

x_1	x_2	x_3
10	20	30

$1/6$	$1/3$	$1/2$
-------	-------	-------

← This determines probabilistically

GUARANTEE

$$\mathbb{E} \left[\sum_{i=1}^n \|x_i - \mu_{z_i}\|^2 \right]$$

→ Assume initialization done
Randomly

→ over randomness.

→ Ideally, $\min_{z_1, \dots, z_n} \sum_{i=1}^n \|x_i - \mu_{z_i}\|^2$

→ cannot get here but can get pretty close.

$$\mathbb{E} \left[\sum_{i=1}^n \|x_i - \mu_{z_i}\|^2 \right] \leq O(\log k)$$

$$\left[\min_{z_1, \dots, z_n} \sum_{i=1}^n \|x_i - \mu_{z_i}\|^2 \right]$$

23.6 - Choice of K

$$\rightarrow F(z_1, \dots, z_n) = \sum_{i=1}^n \|x_i - \mu_{z_i}\|^2$$

→ How this function can be reduced

→ can choose n (no. of points)

→ need another function

(penalize large values of K)

Ideally (smallest [Objective function
+ Penalty k])

There are 2 criterion for this function.

1) AIC - Akaike Information Criterion

$$[2K - 2 \log[L(\theta^*)]]$$

2) BIC - Bayesian Information Criterion

$$K \log(n) - 2 \log[L(\theta^*)]$$